

PROJECT AURORA: AUTONOMOUS LUNAR ROVER MISSION SPECIFICATION

1. Mission Overview & Power Systems

Project Aurora is an autonomous lunar rover developed by Nebula Aerospace in 2025.

The rover has a total dry mass of 450 kilograms and is engineered to operate in the lunar South Pole region.

Power is provided by a triple-junction gallium arsenide solar array generating 1.2 kilowatts of peak power, paired

2. Navigation and Sensor Payload

Aurora's navigation suite includes a dual 64-beam LiDAR system operating at a wavelength of 905 nanometers,

The onboard compute cluster runs real-time SLAM (Simultaneous Localization and Mapping) algorithms on an e

3. Scientific Drill and Analysis Suite

For subsurface geotechnical exploration, Aurora deploys a rotary-percussive coring drill capable of extracting cylindrical samples up to 10 cm in diameter and 1 m in length. Subsurface samples are transferred into a sealed Raman spectrometer carousel that identifies volatile organic compounds and mineral composition.

4. Communications Architecture

Telemetry and scientific data are transmitted via an X-band high-gain phased array antenna communicating with Earth-orbiting relay satellites. The communication subsystem achieves a peak downlink rate of 25 megabits per second and a continuous uplink rate of 10 megabits per second.

5. Thermal Management & Hibernation Protocols

During the 14-day lunar night, ambient surface temperatures drop to minus 130 degrees Celsius.

Aurora enters a low-power hibernation mode where non-essential subsystems are de-energized.

Critical avionics and battery cells are maintained above minus 20 degrees Celsius using two radioisotope heaters.

Total thermal power dissipation from each RHU is 1.0 watt of continuous heat.

6. Mission Lifetime and Targets

The primary mission duration is planned for 18 Earth months across 12 lunar day-night cycles, with an exploration